TESSERA

Identity for Humans and Their Agents

Working Draft v0.5

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The Roman tessera hospitalis *was a clay tile broken between two strangers. Each carried a half; reuniting the pieces proved identity and established trust across distance.*

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Abstract

Tessera is an open protocol that makes AI agent actions contingent on human-delegated, cryptographically verified permission. It is the permission layer for autonomous agents: before an agent can send an email, execute a transaction, post content, or run a command, it must present a valid Tessera credential proving that a real human authorised it, what it is permitted to do, and when that permission expires.

The protocol anchors identity to real-world financial verification (bank KYC via open banking), issues W3C Verifiable Credentials backed by Semaphore zero-knowledge proofs, and extends trust from humans to agents through scoped, time-limited, revocable delegations signed with Ed25519.

As AI agents proliferate—browsing, transacting, communicating, and making decisions autonomously—the internet faces three converging crises. Platforms cannot distinguish humans from bots. Agents cannot prove who authorised them or what they are permitted to do. And AI-generated content is flooding every medium with no reliable way to verify human authorship. Tessera addresses all three with a single primitive: a privacy-preserving credential that blocks unauthorised action by default.

Tessera is not a cryptocurrency, does not require biometric scanning, and minimises personal data by design. It is an open standard built on W3C Verifiable Credentials (Data Model 2.0), Semaphore zero-knowledge proofs (Groth16), and OID4VC presentation protocols, delivered through a consumer app, an open-source SDK, and agent framework integrations that make permission the default, not the exception.

1. Problem Statement

1.1 The Cost of Fake Identity Is Zero

Creating a digital identity is free. An email address, a social media account, a forum registration—none require meaningful proof of existence. This asymmetry makes Sybil attacks trivially cheap: a single actor can instantiate thousands of fake identities at negligible cost, undermining democratic participation, content moderation, fraud prevention, and market integrity.

Current defences—CAPTCHAs, phone verification, email confirmation—impose friction on legitimate users while remaining easily defeated by automated systems. They test capability, not identity.

1.2 The Agent Identification Gap

The emergence of autonomous AI agents introduces a new dimension to the identity problem. Systems such as OpenClaw, Claude Code, MCP-connected assistants, and open-source agent frameworks can send emails, execute transactions, post content, and interact with services autonomously on behalf of human principals. Yet they have no standard way to prove:

- That a human principal exists and authorised the agent

- What the agent has been permitted to do (scope)
- When that authorisation expires
- How to contact or hold accountable the responsible human

Without this, platforms face an impossible choice: block all automated access (harming legitimate automation) or permit it (opening the door to unbounded bot activity). Neither is acceptable as agentic AI becomes mainstream.

1.3 The Content Provenance Crisis

AI-generated content is flooding every medium. Text, images, music, video, and code can now be produced at scale with no reliable way to distinguish machine output from human creation. This threatens the livelihoods of creators, the integrity of information ecosystems, and the trust that underpins creative economies.

Existing provenance initiatives (C2PA, Adobe Content Credentials) are tied to specific tools—you must use particular software or hardware to generate provenance metadata. There is no tool-agnostic way for a human creator to sign their work and prove "a real, unique person made this" without revealing who they are.

1.4 The Credential Fragmentation Problem

Even where identity verification exists, it is siloed. A user verified on one platform must reverify on every other. There is no portable credential a human can carry that says "I am a verified unique person"—without revealing who they are. The W3C Verifiable Credentials standard [1] exists to solve this, but lacks adoption because no credible issuer has anchored it to a widely-trusted real-world signal.

2. Design Goals

Tessera is designed to satisfy the following properties simultaneously:

Property	Description
Sybil resistance	One real-world person maps to exactly one root credential. Economic cost of multiple identities is prohibitive.
Privacy preservation	No personal data is transmitted during verification. Platforms learn only that the credential is valid, its tier, and (for agents) its scope.
Portability	A credential issued once is valid everywhere that accepts the Tessera standard.
Agent-native	Delegation to autonomous agents is a first-class protocol primitive with scoped permissions and instant revocation.
Human provenance	Creators can sign their work with a zero-knowledge proof of human authorship.
Progressive trust	Multiple verification layers stack to increase credential strength. Higher tiers unlock more capabilities.
Accessibility	No cryptocurrency, no hardware tokens, no biometric devices. Any person with a smartphone can participate.
Open standard	The protocol is open, auditable, and not controlled by any single entity.
Economic sustainability	Revenue from verification fees and agent transaction stamps funds a human dividend, aligning incentives without a cryptocurrency token.

Table 1: Tessera design goals.

3. Protocol Architecture

Tessera is structured as four protocol layers, each buildable independently and composable into the full system.

3.1 Layer 1—The Anchor (Root of Trust)

The anchor is the real-world signal that makes a Tessera credential meaningful. Multiple anchor types are supported, ranked by assurance level:

Tier	Method	Description
1	Bank KYC	Open banking verification (Plaid, TrueLayer). Strongest assurance. Account must be real, active, and not previously used.
2	Card verification	Stripe-style microcharge with name match. Near-global coverage.
3	Phone + email	Mobile carrier verification combined with email confirmation.
4	Verification network	Existing verified users attest to new users. Graph-weighted trust scoring.

Table 2: Anchor tiers ranked by assurance level.

The anchor tier is encoded in the issued credential, allowing platforms to set minimum assurance levels. A social network might accept Tier 4; a financial service might require

Tier 1. Critically, no anchor data is stored after credential issuance. Only a one-way hash of the anchor identifier is retained for deduplication.

Tiers are additive: a user who starts with Tier 4 (verification network) can upgrade to Tier 1 (bank KYC) at any time, and both verification signals are reflected in their credential. More layers of verification yield a stronger credential with access to more features across the ecosystem.

3.2 Layer 2—The Credential

Upon successful anchor verification, the Tessera issuer generates a Verifiable Credential conforming to the W3C VC Data Model [1], backed by a Semaphore [4] zero-knowledge proof:

```
{
  "type": "TesseraHumanCredential",
  "version": "1.0",
  "claims": {
    "isUniqueHuman": true,
    "anchorTier": 1,
    "anchorJurisdiction": "EU",
    "issuedAt": 1743000000,
    "expiresAt": 1774536000
  },
  "proof": "<semaphore-zk-proof>"
}
```

No name, email, account number, or personally identifiable information is included. The zero-knowledge proof (Groth16 via Semaphore) allows any verifier to confirm the credential is valid without learning anything about the underlying anchor. Credentials are stored locally on the user's device. The issuer retains only a nullifier registry—a set of hashed credential identifiers used to detect double-presentation.

3.3 Layer 3—Agent Delegation

A human credential holder can delegate to one or more agent identities. Each agent credential is cryptographically signed by the parent human, referencing their identity commitment and encoding a scope and expiry:

```
{
  "type": "TesseraAgentCredential",
  "parentCommitment": "<human-identity-commitment>",
  "agentName": "my-agent",
  "scope": {
    "canPost": true,
```

```
    "canTransact": true,
    "maxTransactionValue": 50,
    "currency": "EUR",
    "allowedCategories": ["saas", "api"]
  },
  "issuedAt": 1743000000,
  "expiresAt": 1743604800,
  "parentSignature": "<ed25519-signature>"
}
```

Delegations form a tree with the following properties:

1. A child credential cannot exceed the scope of its parent.
2. Revoking a parent immediately invalidates all children.
3. The human holds the revocation key; no third party can revoke on their behalf.
4. Agents can further sub-delegate within their own scope.

3.4 Layer 4—Verification API

Platforms and services integrate Tessera through an open-source SDK. The verification flow completes in under 200 ms:

1. Platform requests credential from user or agent.
2. User or agent presents signed Verifiable Credential.
3. SDK verifies ZK proof via Semaphore (cryptographic validity).
4. SDK verifies issuer signature (credential authenticity).
5. SDK checks nullifier registry (double-presentation detection).
6. SDK checks revocation status (for agent credentials).
7. Returns: `{ valid, type, tier, scope }`.

4. Standards Alignment

Tessera is designed to interoperate with the emerging digital identity stack rather than compete with it. The protocol builds on established standards and prepares compatibility paths with evolving regulatory frameworks.

4.1 Core Standards

Standard	How Tessera uses it
W3C VC Data Model 2.0	Tessera credentials conform to the VC Data Model 2.0, which reached W3C Recommendation status in May 2025. This ensures credential portability across any verifier that supports the standard.
Semaphore V4 / Groth16	Anonymous group membership proofs with nullifier-based double-signal prevention. Semaphore V4 is maintained by Privacy & Scaling Explorations (Ethereum Foundation) with a completed trusted setup (400+ participants) and production deployment through World (17.9M verified users). Groth16 produces 192-byte proofs with verification in under 10 ms.
Ed25519	All delegation signatures use Ed25519 via the Web Crypto API where available. Browser-generated keys use the `extractable: false` flag to prevent accidental exfiltration.
OID4VC	The credential presentation layer is designed for compatibility with OpenID for Verifiable Credentials (OID4VCI / OID4VP), which is becoming the implementation centre of gravity for VC ecosystems including the EU Digital Identity Wallet.

Table 3: Core standards used by Tessera.

4.2 Regulatory Compatibility

Tessera is designed with awareness of the emerging regulatory landscape for digital identity and AI agents, though no dedicated statutory regime for AI-agent identity currently exists.

Framework	Tessera alignment
EU eIDAS 2.0	The European Digital Identity Wallet framework entered into force in 2024, with member states required to offer at least one wallet by end of 2026. EUDI architecture references W3C VC Data Model 2.0 and SD-JWT VC formats. Tessera credentials are designed for future interoperability with EUDI-compatible wallets, adding agent delegation on top of the human credential.
UK DVS Trust Framework	The UK Digital Verification Services trust framework 1.0 was announced in March 2026, with a live certification register. Tessera's architecture is compatible with DVS certification requirements, providing a path to regulatory credibility for UK deployments.
EU AI Act (Article 50)	Transparency obligations for AI-generated content become applicable in August 2026. Tessera's human provenance watermark and agent credential system provide infrastructure for content labelling compliance.
GDPR / UK Data Protection	Tessera's use of zero-knowledge proofs, nullifiers, and hashed identifiers supports data minimisation and security by design. However, pseudonymised data may still constitute personal data under ICO and EDPB guidance where re-identification remains reasonably possible. Tessera does not claim exemption from data protection law; it claims strong compliance posture through architectural minimisation.

Table 4: Regulatory framework compatibility.

Tessera does not directly access payment-account data or initiate payments, which is intended to avoid the need for Account Information Service Provider (AISP) or Payment Initiation Service Provider (PISP) authorisation under UK/EU payment services regulations. Instead, Tessera consumes verification results from already-authorised open banking providers such as TrueLayer. This architectural choice is deliberate and should be validated by legal counsel before production deployment.

4.3 Standards Participation

Tessera intends to participate in the following standards bodies and working groups as the protocol matures:

- **OpenID Foundation AIIM Community Group** — created in 2025 to address AI identity management, and the most directly relevant forum for Tessera's agent delegation model.
- **NIST AI Agent Standards Initiative** — launched in February 2026 with explicit focus on agent security and identity.
- **W3C VC Working Group** — active through 2028, providing the credential substrate Tessera builds on.

- **FIDO Alliance Digital Credentials Working Group** — launched December 2025, relevant where passkeys and verifiable credentials converge.

5. The Tessera App

The protocol is delivered through a consumer application—the Tessera App—that serves as both a credential wallet and the primary interface for identity management. The app is designed to make zero-knowledge identity as intuitive as a passport.

5.1 Tessera Passport

Each user's primary view is their Tessera Passport: a digital identity card displaying their verification tier, the types of verification they have completed, and a QR code for instant sharing. The passport carries no personal information—it is a visual representation of a cryptographic credential.

Users can progressively strengthen their passport by completing additional verification layers. Starting from a social verification (Tier 4), they can add phone confirmation (Tier 3), card verification (Tier 2), and bank KYC (Tier 1). Each layer adds credential strength and unlocks additional capabilities across the Tessera ecosystem.

5.2 Agent Wallets

The app provides a dedicated interface for creating and managing agent credentials. A user can:

- Create a new agent wallet with a human-readable name
- Define the agent's scope (what it can do, spending limits, permitted categories)
- Set an expiry (hours, days, or weeks)
- Export a credential token for use in agent configurations (Claude Code, OpenClaw, MCP servers, or any compatible framework)
- Monitor agent activity in a structured log
- Revoke any agent credential instantly with a single action

This is the first consumer-facing identity and wallet system for AI agents—allowing any user to manage, authorise, and control the agents acting on their behalf.

5.3 Verification Network

The app includes a verification network through which users strengthen their own credentials and those of others. When two users verify each other (in person via NFC or remotely via the app), both credentials gain trust weight.

The network uses graph-weighted trust scoring rather than simple referral counting. Vouches from users in different social clusters carry more weight than vouches from within a tightly-connected group, providing natural resistance to collusion rings. The system uses institutional framing ("verification network") rather than social framing ("friends" or "followers"), reflecting that Tessera is an identity infrastructure, not a social network.

6. Human Provenance

As AI-generated content floods every digital medium, the ability to prove human authorship becomes critical infrastructure. Tessera provides tool-agnostic content provenance through zero-knowledge human attestation.

6.1 The Problem

AI systems can now generate text, images, music, video, and code that is indistinguishable from human-created work. This threatens creative economies (streaming platforms flooded with AI-generated music), information integrity (AI-generated articles and social media posts), and trust in digital content broadly.

Existing provenance solutions (C2PA, Adobe Content Credentials) require specific creation tools. A photographer must use a C2PA-compatible camera; a writer must use a participating text editor. There is no way for a creator using any tool to independently attest "I, a verified unique human, created this work."

6.2 Tessera Watermark

A Tessera credential holder can generate a **Tessera Watermark**—a zero-knowledge signature that proves:

- The signer is a verified unique human (Semaphore group membership proof)
- The signature is bound to a specific content hash (the work cannot be swapped)
- The signer's identity is not revealed (zero-knowledge property)

The watermark is a compact data structure appended to or embedded in the content. Any verifier can confirm "a real, verified human signed this" without learning who. The same human can watermark multiple works without those works being linkable to each other or to the creator's identity.

6.3 Applications

- **Music and audio:** Streaming platforms can distinguish human-composed tracks from AI-generated content for curation, playlist placement, and royalty eligibility.

- **Written content:** Publishers and platforms can verify human authorship of articles, reviews, and academic work.
- **Visual art:** Marketplaces can flag or prioritise human-created artwork.
- **Code:** Open-source projects can verify that contributions come from real humans, not automated systems.
- **Social media:** Platforms can surface human-created posts with higher confidence.

The watermark system builds directly on the existing Tessera credential infrastructure—no new cryptographic primitives are required. A user who already has a Tessera Passport can watermark content from within the app or via the SDK.

7. Ecosystem

Tessera is designed as a platform that third-party applications and services can build upon. The ecosystem model is inspired by the integration patterns of open identity standards (OAuth, OpenID Connect) but extended to support agent credentials and human provenance.

7.1 Integration Model

Any application can integrate Tessera through the open-source SDK. Integration surfaces include:

- **"Verify with Tessera" button:** A web or app-based verification flow, analogous to "Sign in with Google" but privacy-preserving. The platform learns only the verification tier and (if agent) the scope—never the user's identity.
- **Agent verification endpoint:** An API endpoint that AI agents call to present their delegation credentials. MCP servers, API gateways, and agent frameworks can verify agent authorisation in a single HTTP call.
- **Watermark verification:** Content platforms can check Tessera Watermarks to confirm human authorship, displayed as a "Human Verified" badge or used for algorithmic curation.

7.2 Ecosystem Partners

The initial ecosystem targets two verticals:

Agent-native services. MCP servers, AI platforms, and automation tools integrate Tessera to verify that connecting agents are authorised by real humans. This addresses the immediate pain point of bot proliferation in agent ecosystems.

Human provenance services. Content platforms, creative marketplaces, streaming services, and publishers integrate Tessera Watermarks to distinguish human-created content

from AI-generated material.

Agent framework integrations. The initial integration target is OpenClaw, the open-source agent framework with over 300,000 users and 13,000+ community-built skills. OpenClaw agents already emit approval events for sensitive actions; Tessera Guard replaces the "ask the human every time" pattern with pre-delegated, revocable authority. Subsequent integrations target Claude Code, MCP-compatible tools, and LangChain-based agent systems.

These verticals are complementary: the agent vertical proves "this agent is authorised by a real human," while the provenance vertical proves "this content was created by a real human." Both stem from the same underlying credential.

7.3 The Guard Pattern

The primary integration model for agent frameworks is the **Tessera Guard**—a middleware layer that intercepts sensitive agent actions and blocks them unless a valid Tessera credential authorises the operation.

The guard pattern inverts the default. Instead of agents acting freely unless stopped, agents cannot act unless explicitly permitted. This transforms Tessera from an optional verification service into a required permission layer.

How it works

The guard sits between the agent runtime and its tools. When the agent invokes a sensitive action (sending email, executing a payment, running a shell command, posting content), the guard intercepts the call and evaluates it against the bound Tessera credential:

```
function onToolInvocation(action, context) {
  if (!isSensitive(action)) return allow();

  const credential = context.agent.tesseraCredential;
  const result = tessera.verify(credential, {
    action: action.type,
    resource: action.estimate
  });

  if (!result.valid) return deny("No valid credential");
  if (!result.scope_ok) return deny("Outside delegated scope");
  return allow();
}
```

The verification call checks credential validity, issuer signature, scope containment, expiry, and revocation status. If any check fails, the action is blocked and the agent receives a structured explanation of why—enabling it to communicate the limitation to the user within

the conversation.

Protected action classes

Action class	Risk
`message.send`	Agent sends messages (email, chat, SMS) on behalf of the human. Risk: impersonation, spam, social engineering.
`payment.intent`	Agent initiates purchases or transfers. Risk: unauthorised spending, financial fraud.
`exec.shell`	Agent executes system commands. Risk: data exfiltration, system compromise, privilege escalation.
`content.publish`	Agent posts content publicly. Risk: reputational damage, misinformation, legal liability.

Table 5: Protected action classes in the Tessera Guard.

Agent-side legibility

When the guard denies an action, the agent receives a structured response that it can interpret and communicate naturally. Rather than failing silently, the agent can tell the user:

> "I'm not authorised to send emails to more than 20 recipients. My current Tessera credential limits bulk messaging. You can issue a broader credential from the Tessera app if you'd like me to proceed."

This makes the permission system legible inside the agent conversation itself, turning security enforcement into a natural part of the agent interaction rather than an opaque error.

Instant revocation

If a human revokes an agent's Tessera credential (via the app, with a single tap), the next sensitive action the agent attempts will fail immediately. The guard queries the issuer's revocation status on every verification call. There is no propagation delay—revocation is effective within one verification round-trip (typically under 200 ms).

This provides a "kill switch" for agent behaviour that works across all platforms and tools simultaneously, without requiring the human to individually disconnect the agent from each service.

8. Economic Model

Tessera sustains itself through a three-layer economic model that aligns incentives between platforms, agent operators, and verified humans. Critically, Tessera does not issue a cryp-

tocurrency token. All value flows in fiat currency through standard payment processors.

8.1 Platform Verification Fees

Platforms that integrate the Tessera verification API pay a per-verification fee. This is the core revenue mechanism, analogous to Stripe charging per transaction or Twilio charging per SMS. Platforms pay because the alternative—building proprietary Sybil resistance, KYC infrastructure, and agent verification—costs orders of magnitude more.

Pricing is volume-tiered, starting at approximately £0.01–0.05 per verification. The open-source SDK remains free; the fee applies only to the hosted verification service (nullifier registry, issuer signature validation, and group root management).

8.2 Agent Transaction Stamps

Every scoped agent action—a purchase, a post, a message sent on behalf of a human—can optionally require a **transaction stamp**. The stamp is a micropayment (typically £0.001–0.01) paid by the agent's human owner and recorded alongside the verification result.

Transaction stamps serve two purposes. First, they create economic friction that makes automated abuse expensive. A bot army performing 10,000 actions costs £10–100 in stamps—trivial for a legitimate user performing a few actions per day, but prohibitive for a spammer. This is economically equivalent to the Hashcash proof-of-work concept applied at the identity layer rather than the network layer.

Second, stamps generate revenue that scales with agent activity rather than user count. As agents become more capable and perform more actions, the stamp revenue grows proportionally, creating a natural alignment between protocol utility and protocol sustainability.

Stamps are optional. Platforms choose whether to require them and can set their own thresholds. The protocol defines the stamp mechanism; platforms define the policy.

Direct per-action card charges at these amounts are below the minimum transaction thresholds of major payment processors (for example, Stripe's minimum charge is £0.30 GBP). In practice, stamps will be aggregated into periodic billing cycles or prepaid balance drawdowns rather than charged individually. The economic friction effect is preserved; the payment-rail constraint is respected.

8.3 Why the Model Compounds

The three layers reinforce each other. Verification fees monetise trust at the platform edge. Transaction stamps monetise action inside agent workflows. The human dividend turns verification from a pure cost centre into a shared upside mechanism. This creates a flywheel: more useful agents increase platform demand for verification; more platform demand increases protocol revenue; more revenue increases the dividend; a larger dividend increases the incentive

for real humans to verify and remain active in the ecosystem.

This is structurally different from advertising-driven social networks, where users create value but rarely participate directly in the upside. It is also different from token-governed identity systems, where economic value is mediated through speculative assets. Tessera instead treats identity as infrastructure and distributes a share of infrastructure revenue back to the humans whose verified uniqueness makes the system possible.

8.4 Human Dividend

A fixed percentage of gross revenue from verification fees and transaction stamps is redistributed to verified humans. This is the **human dividend**—a direct fiat payment to every Tier 1 verified user, distributed monthly.

The mechanism is simple: 30% of gross protocol revenue enters the dividend pool. The pool is divided equally among all Tier 1 verified credential holders at the end of each distribution period. The per-user amount is small at early scale and grows with adoption.

The human dividend differs from token-based redistribution (as in Worldcoin) in three important ways:

1. There is no speculative token. The dividend is denominated in fiat currency and paid through standard payment rails.
2. The amount is deterministic and transparent. Users can calculate their expected dividend from publicly reported revenue figures.
3. The dividend grows with real economic activity (agent transactions, platform verifications), not with speculative market dynamics.

This model creates a direct incentive for verification: every verified human earns a share of the value generated by the agent economy they are enabling. The more useful agents become, the more verified humans earn.

8.5 Tessera Does Not Touch Money

Tessera is identity infrastructure, not financial infrastructure. Spending limits on agent delegations are permissions, not balances. When an agent credential specifies `maxTransactionValue: 50 EUR`, the platform receiving the credential enforces the limit through its own payment system. Tessera verifies who is authorised and what they are authorised to do. The platform handles how money moves.

This design avoids payment institution regulation (FCA in the UK, PSD2 in the EU). Verification fees and stamp payments are processed through standard payment processors (Stripe). The human dividend is distributed as revenue sharing, not as stored value.

9. Comparison with Existing Approaches

System	Sybil resist.	No biometrics	Agent support	Human proven.	Open standard
Email / phone	Low	Yes	No	No	Partial
Google / Apple SSO	Medium	Yes	No	No	No
Worldcoin / World ID	High	No	No	No	Partial
Gitcoin Passport	Medium	Yes	No	No	Yes
C2PA / Content Cred.	N/A	Yes	No	Partial	Yes
Microsoft Entra Agent ID	N/A	Yes	Native	No	No
Okta / Auth0 for Agents	N/A	Yes	Native	No	No
Tessera	**High**	**Yes**	**Yes**	**Yes**	**Yes**

Table 6: Comparison with existing identity and provenance systems.

Tessera occupies a unique position: high Sybil resistance without biometrics, native agent support, content provenance, and an open standard. No existing system combines all five properties.

Relative to Worldcoin specifically, Tessera differs in three fundamental ways: it does not require biometric hardware (no orb), it does not issue or depend on a cryptocurrency token, and it provides first-class support for AI agent delegation. Tessera shares Worldcoin's ambition of universal digital identity but pursues it through existing financial infrastructure (bank KYC) and social trust (verification networks) rather than biometric scanning.

10. Privacy Guarantees

Party	Learns	Does Not Learn
Verifying platform	Credential is valid, anchor tier, credential age, agent scope (if agent)	User's name, bank, email, or any PII
Tessera issuer	Anchor hash at issuance (discarded after), nullifier at re-verification	What platforms the user verifies with
Tessera issuer (recovery)	SHA-256 hashes of optional recovery contacts; encrypted backup blob	Plaintext email, phone number, or backup contents
Content consumers	That a verified human signed the content	Who the creator is, or whether two works share a creator
Other users	Nothing—credentials are not publicly visible	—
Governments	Nothing without legal process directed at the user's bank	—

Table 7: Privacy guarantees by party.

Tessera's privacy model is designed to be strictly stronger than existing verification systems (phone verification, social login, document KYC), all of which expose significant PII to verifying parties. The content provenance system adds a new privacy guarantee: creators can prove human authorship without revealing their identity, and multiple works by the same creator cannot be linked.

Data protection note. Tessera's use of hashed identifiers, nullifiers, and zero-knowledge proofs supports data minimisation by design. However, pseudonymised data may still constitute personal data under UK ICO and EU EDPB guidance where re-identification remains reasonably possible through auxiliary information. Tessera does not claim exemption from GDPR or the UK Data Protection Act; it claims a strong compliance posture through architectural data minimisation, purpose limitation, and the absence of stored plaintext PII.

Users who provide recovery contacts accept a limited reduction in anonymity: the issuer can route a verification code to the correct email address when recovery is initiated, but cannot read the email address from the stored hash. This is an opt-in convenience feature, not a protocol requirement.

11. Authentication and Recovery

11.1 Credential-as-Authentication

Tessera has no login system, no accounts, no usernames, and no passwords. The credential itself is the authentication. Possession of a valid Tessera credential, protected by device-level

biometrics (FaceID, TouchID, or equivalent), is sufficient to prove identity to any platform.

This model eliminates an entire class of attacks—phishing, credential stuffing, password reuse—because there are no credentials to steal in the traditional sense. The Semaphore identity secret is stored in the device's secure enclave and never transmitted. Verification is performed via zero-knowledge proof, which proves possession of the secret without revealing it.

11.2 Device Security

The app is protected by device-level biometric authentication on every open. This is a local check performed by the operating system (iOS Keychain, Android Keystore) and does not involve Tessera's servers. The biometric data never leaves the device and is not part of the Tessera protocol.

In practice, this means Tessera inherits the security posture of the device platform rather than attempting to replace it. Apple and Google already operate mature secure storage and biometric gating primitives; Tessera uses those primitives as the user-facing access control layer, while keeping the protocol itself focused on cryptographic proof and delegated authority.

11.3 Recovery

Loss of a device means loss of the credential, which must be recoverable without compromising privacy. Tessera supports optional recovery contacts: the user may provide an email address and/or phone number during onboarding.

These contacts are hashed locally (SHA-256) before transmission. The issuer stores only the hashes alongside an encrypted backup blob (the credential and identity secret, encrypted with a key derived from the Semaphore identity). The issuer cannot read the email, cannot read the phone number, and cannot decrypt the backup.

Recovery proceeds as follows:

1. User enters their recovery email on a new device.
2. The app hashes it locally and sends the hash to the issuer.
3. The issuer matches the hash and sends a verification code to the email.
4. The user enters the code, receiving the encrypted backup.
5. The user re-verifies with their bank (TrueLayer) to prove they are the same person.
6. The issuer re-issues the credential with a new identity commitment linked to the original anchor.

Recovery contacts are optional. Users who decline accept the risk that device loss requires full re-verification. This is an explicit privacy-convenience tradeoff that the user controls.

This recovery model preserves the core protocol invariant: recovery should restore access, not

create a second copy of the credential. Re-issuance invalidates the lost device's credential path and replaces it with a new identity commitment bound to the same verified anchor, preventing recovery from becoming a duplication vector for Sybil attacks.

12. Threat Model

Attack	**Defence**
Sybil (multiple fake identities)	Each bank account maps to one credential. Social vouching uses graph-weighted trust scoring with collusion resistance.
Stolen credential reuse	Credentials are device-bound. Nullifiers prevent double-presentation per platform.
Agent impersonation	Agent credentials are Ed25519-signed by the parent human. Forgery requires the parent's private key.
Scope escalation	Scope is encoded in the delegation and verified by the SDK. Scope containment is enforced cryptographically.
Collusion rings	Graph-weighted trust scoring penalises vouches from within tightly-connected clusters. Cross-cluster vouches carry more weight.
False provenance	Watermarks are bound to content hashes. Swapping the content invalidates the watermark.
Platform fingerprinting	Zero-knowledge proofs ensure platforms learn no information beyond credential validity.
Credential resale	Credentials are non-transferable. The ZK proof is tied to a device key generated at issuance.

Table 8: Threat model and defences.

Tessera does not claim to prevent all misuse by verified humans. A real person can still behave badly after verification. The progressive verification model creates layered accountability: higher-tier credentials carry more trust weight and unlock more capabilities, creating natural incentives for genuine verification.

13. Open Questions

Question	Current Thinking
Cross-anchor deduplication	A person with accounts at multiple banks could obtain multiple credentials. Mitigation requires a shared identifier across anchors, which varies by jurisdiction.
Nullifier registry scaling	SQLite is suitable for local, edge, and low-to-medium traffic deployments. A global nullifier authority with high write concurrency will require migration to a multi-writer store (e.g. PostgreSQL). Bitstring status lists (W3C) are the preferred revocation pattern at scale.
Mobile key storage	iOS Secure Enclave supports P-256 but not Ed25519. Hardware-backed Ed25519 on iOS requires storing a wrapped key in Keychain rather than native enclave operations. Some authentication surfaces may adopt WebAuthn/passkeys (P-256) for maximum hardware assurance.
Micropayment batching	Agent transaction stamps at £0.001–0.01 are below card-rail minimums. Implementation requires prepaid balances, periodic billing, or streaming payment abstractions rather than direct per-action charges.
Digital ID certification	UK DIATF and EU eIDAS 2.0 certification paths for real-world identity acceptance. DVS certification takes approximately 1–2 weeks once engaged with a conformity assessment body.
Agent credential standards	No existing standard for agent authorisation credentials. The OpenID Foundation AIIM Community Group and NIST AI Agent Standards Initiative are the most relevant forums. Tessera may propose a W3C VC extension for agentic scope encoding.

Table 9: Open design questions.

14. Roadmap

Phase	Scope
Phase 0—Foundation (complete)	Protocol specification and reference implementation. SDK with Semaphore ZK proofs, Ed25519 signatures, and SQLite nullifier registry. Published as `@tessera-protocol/sdk` on npm. Credential wallet app (web + iOS). CLI demo. Issuer service. 20-page whitepaper.
Phase 1—OpenClaw integration	Tessera Guard middleware for OpenClaw. Permission enforcement on email, payments, shell, and content actions. Agent-side permission legibility. Revocation integration. First real users from the OpenClaw community.
Phase 2—Bank anchor	TrueLayer open banking sandbox integration for Tier 1 credentials. App onboarding flow with recovery contacts. Real bank-verified credentials replacing demo mode.
Phase 3—Ecosystem expansion	MCP server for Claude Code and Claude Desktop. Tessera-signed skill manifests for OpenClaw. Micropayment infrastructure (agent transaction stamps). Human dividend tracking. Migrate nullifier registry to PostgreSQL for production scale. Adopt W3C Bitstring Status List for credential revocation.
Phase 4—Scale	Full W3C VC compliance. Content watermark SDK. Apple Wallet pass. Multi-issuer support. Enterprise tier. Security audit.
Phase 5—Institutional	UK DIATF certification exploration. Nullifier registry decentralisation. Foundation-style governance. Join OIDF AIIM Community Group and NIST AI Agent Standards Initiative. Pursue UK DVS trust framework certification.

Table 10: Implementation roadmap.

15. Open Source and Governance

Tessera is committed to an open core model. The following components are released under Apache 2.0:

- The Tessera credential schema and protocol specification
- The verification SDK (`@tessera-protocol/sdk`)
- The ZK proof library (built on Semaphore)
- The watermark generation and verification library
- Reference implementations of agent delegation

The following components are developed as managed services:

- The Tessera App
- The hosted nullifier registry

- The verification network infrastructure
- Enterprise integrations and support

Protocol governance in Phase 1 rests with the founding team. An advisory board of identity researchers and security engineers will be established before the v1.0 release. The long-term goal is a foundation model with multi-stakeholder governance, similar to the W3C or OpenID Foundation.

16. Get Involved

Tessera is in its earliest stage. We are actively seeking:

- Feedback on the protocol design from identity and cryptography researchers
- Early ecosystem integrators (MCP servers, agent platforms, content platforms)
- Contributors to the open-source implementation
- Advisors with experience in ZK cryptography, open banking, digital identity regulation, or agent systems

GitHub: https://github.com/tessera-protocol/tessera

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